Solid State Devices EE3001 Quiz 1 Syllabus

Semiconductor Properties

A semiconductor in its pure form is called an intrinsic semiconductor. In an
intrinsic semiconductor, the number of free electrons is equal to the number of holes, as they are generated simultaneously by the breaking of bonds. A doped semiconductor is called an extrinsic semiconductor. If the dopant in an extrinsic semiconductor is a donor, free electrons are created without simultaneous creation of holes. Consequently, the number of electrons will be more than the number of holes and the semiconductor is
said to be n-type (n stands for negatively charged electrons). Alternately, if the dopant is an acceptor, there will be more holes than electrons and the semiconductor is said to be ptype (p stands for positively charged holes).

Since electrons and holes are created in pairs, the electron concentration
in the conduction band (n) is equal to the hole concentration in the valence band (p), i.e.
n = p = n_i where n_i is called the intrinsic carrier concentration. n_i increases with rise in temperature (more EHPs).

$\sigma = nq\mu_n + pq\mu_p$ (maximum resistivity of Si is when sample is slighly p-type). $\mu \propto T^{-\frac{3}{2}}$ due to lattice scattering and $\mu \propto \frac{T^{\frac{3}{2}}}{N}$ due to impurity scattering

Electron and hole densities in equilibrium

effective mass $m^* = \frac{\hbar^2}{\left(\frac{d^2E}{dk^2}\right)}$

the number of states in the conduction band with energies between E and E+dE is given by

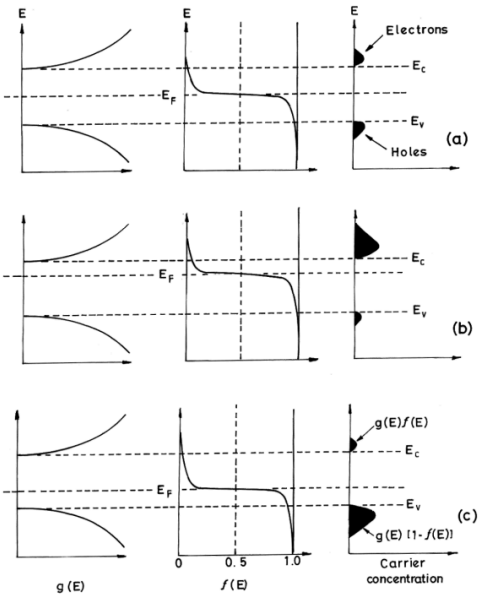
$$g(E)dE = 4\pi \left(\frac{2m_e}{h^2}\right)^{3/2} (E - E_c)^{1/2} dE \quad \text{for } E > E_c$$

the number of states in the valence band with energies between E and E+dE is

$$g(E)dE = 4\pi \left(\frac{2m_h}{h^2}\right)^{3/2} (E_v - E)^{1/2} dE \quad \text{for } E < E_v$$

The other result that we shall assume is the Fermi-Dirac distribution function.
Electrons in solids follow Fermi-Dirac statistics, which can be expressed simply as follows: the probability f(E) that an electron occupies an available state at an energy level (E) at thermal equilibrium at a temperature T(K) is

$$f(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{kT}\right)}$$



$$n = \int_{E_C}^{E_{top}} f(E)g(E)dE = N_C \exp\left[-\left(\frac{E_C - E_F}{kT}\right)\right]$$

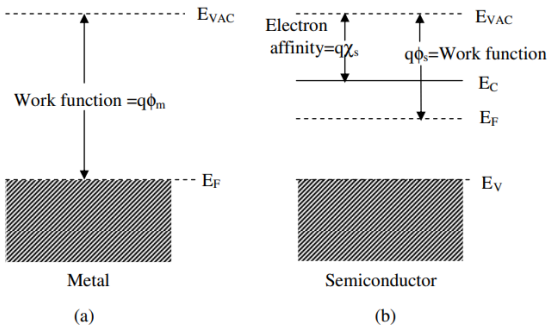
where $N_C = 2\left(\frac{2\pi kTm_e^*}{h^2}\right)^{3/2}$ is called the effective density of states in the conduction band.

Similarly, $p = N_V \exp\left[-\left(\frac{E_F - E_V}{kT}\right)\right]$ where $N_V = 2\left(\frac{2\pi kTm_h^*}{h^2}\right)^{3/2}$

E_i denotes the position of the Fermi energy level in an intrinsic semiconductor, $E_i = \frac{E_c+E_v}{2} - \frac{KT}{2} ln \frac{N_c}{N_v}$ and $n_i = \sqrt{N_C N_V} \exp\left[-\left(\frac{E_g}{2kT}\right)\right]$.

Charge Neutrality: $N_A + n = N_D + p$, n-type means $n = N_D$ and p-type means $p = N_A$. $n.p = n_i^2$ is always valid.

No discontinuity or gradient can exist in the Fermi Energy Level at thermal equilibrium



Quasi Fermi Level / IMREF

$n = n_i \exp\left(\frac{E_{Fn}-E_i}{kT}\right); p = n_i \exp\left(\frac{E_i-E_{Fp}}{kT}\right); np = n_i^2 \exp\left(\frac{E_{Fn}-E_{Fp}}{kT}\right)$

Poisson, Drift Diffusion and Continuity Equations (Fundamentals)

$J_n = J_{ndr} + J_{ndiff} = qn\mu_n\mathcal{E} + qD_n\frac{dn}{dx} = n\mu_n\frac{d(E_{Fn})}{dx}$

$J_p = J_{pdr} + J_{pdiff} = qp\mu_p\mathcal{E} - qD_p\frac{dp}{dx}p\mu_p\frac{d(E_{Fp})}{dx}$

$\frac{\partial n}{\partial t} = \frac{1}{q}\frac{\partial J_n}{\partial x} + (G_n - R_n)$

$\frac{\partial p}{\partial t} = -\frac{1}{q}\frac{\partial J_p}{\partial x} + (G_p - R_p)$

$\frac{d\mathcal{E}}{dx} = \frac{\rho}{\epsilon_s} = \frac{q}{\epsilon_s}(p - n + N_D^+ - N_A^-)$

$\frac{d^2\phi}{dx^2} = -\frac{\rho}{\epsilon_s} = -\frac{q}{\epsilon_s}(p - n + N_D^+ - N_A^-)$

D_p and D_n are diffusion constants with $\frac{D_p}{\mu_p} = \frac{D_n}{\mu_n} = \frac{kT}{q} = V_T$ (Einstein's relationship)

Generation and Recombination, Lifetime

Minority carrier hole continuity equation (uniform injection):

$$\frac{dp_n}{dt} = -\frac{1}{\cancel{q}}\frac{\cancel{\partial J_p}}{\partial x} + G_p - R_p + G_L \implies \frac{dp_n}{dt} = G_p - R_p + G_L$$

Under Thermal Equilibrium:

$$G_p = R_{p0} = \beta n_{n0}p_{n0} \quad \text{and} \quad J_p = 0$$

$$R \propto np \implies R = \beta np$$

Rate of change of excess carriers:

$$\frac{d(p_{n0} + \Delta p_n)}{dt} = \frac{\cancel{dp_{n0}}}{\cancel{dt}} + \frac{d(\Delta p_n)}{dt} = \frac{d(\Delta p_n)}{dt}$$

$$\frac{d(\Delta p_n)}{dt} = \beta n_{n0}p_{n0} - \beta(n_{n0} + \Delta p_n)(p_{n0} + \Delta p_n) + G_L$$

$$\frac{d(\Delta p_n)}{dt} = G_L - \beta(\Delta p_n p_{n0} + \Delta p_n n_{n0} + \Delta p_n^2)$$

Applying Low-Level Injection ($\Delta p_n \ll n_{n0}$ and $\Delta p_n^2 \approx 0$):

$$\frac{d(\Delta p_n)}{dt} = G_L - \beta n_{n0}\Delta p_n = G_L - \frac{\Delta p_n}{\tau_p}$$

Where lifetime $\tau_p = \dfrac{1}{\beta n_{n0}}$

Solving the Differential Equation: $\dfrac{d(\Delta p_n)}{dt} + \dfrac{\Delta p_n}{\tau_p} = G_L$

$$\int \frac{d(\Delta p_n)}{G_L\tau_p - \Delta p_n} = \int \frac{dt}{\tau_p} \implies \ln\left(\frac{\Delta p_n - G_L\tau_p}{\Delta p_{n0} - G_L\tau_p}\right) = -\frac{t}{\tau_p}$$

1. Build-up (Light ON, $t < t_{OFF}$):

$$\Delta p_n(t) = G_L\tau_p\left(1 - e^{-t/\tau_p}\right)$$

2. Decay (Light OFF, $t > t_{OFF}$):

$$\Delta p_n(t) = \Delta p_n(t_{OFF})e^{-(t - t_{OFF})/\tau_p}$$

Diffusion Length

— 1. Condition for Steady State Diffusion —

In an n-type semiconductor under illumination, total current $J = J_n + J_p = 0$.

Summing drift and diffusion components for electrons and holes:

$$\epsilon(q\mu_n n + q\mu_p p) + qD_n\frac{\partial n'}{\partial x} - qD_p\frac{\partial p'}{\partial x} = 0$$

Using Quasi-Neutrality: $n' \approx p'$ and $\dfrac{\partial n'}{\partial x} \approx \dfrac{\partial p'}{\partial x}$.

— 2. Minority Carrier Drift Neglect —

From the ratio of diffusion to drift current for holes:

$$\frac{J_{p,\text{diff}}}{J_{p,\text{drift}}} = \frac{p(D_n - D_p) + nD_n}{p(D_n - D_p)} \approx \frac{nD_n}{p(D_n - D_p)}$$Since $n_n \gg p_n,$ hole drift is negligible: $J_p \approx -qD_p\dfrac{\partial p_n}{\partial x}$.

— 3. Simplified Continuity Equation —

At steady state $\left(\dfrac{\partial p}{\partial t} = 0\right)$ and zero bulk generation $(G_p = 0)$:

$$D_p\frac{\partial^2 p_n}{\partial x^2} - \frac{p_n - p_{n0}}{\tau_p} = 0$$— 4. Boundary Conditions —

General solution: $p_n - p_{n0} = A\exp\left(\dfrac{x}{\sqrt{D_p\tau_p}}\right) + B\exp\left(-\dfrac{x}{\sqrt{D_p\tau_p}}\right)$

1. At $x = 0, \quad p_n = p_n(0)$

2. At $x = \infty, \quad p_n = p_{n0}$ (Finite concentration)

— 5. Concentration Profile and Diffusion Length —

Applying BCs, $A = 0$ and $B = [p_n(0) - p_{n0}]$. Let $L_p = \sqrt{D_p\tau_p}$:

$$p_n(x) - p_{n0} = [p_n(0) - p_{n0}]\exp\left(-\frac{x}{L_p}\right)$$ — 6. Hole Diffusion Current —

$$J_{p,\text{diff}} = -qD_p\frac{\partial p_n}{\partial x} = \frac{qD_p}{L_p}[p_n(0) - p_{n0}]\exp\left(-\frac{x}{L_p}\right)$$

$$J_{p,\text{diff}}(x) = \frac{qD_p}{L_p}p'(x)$$

L_p is the average distance traveled by the diffusing carriers before they recombine